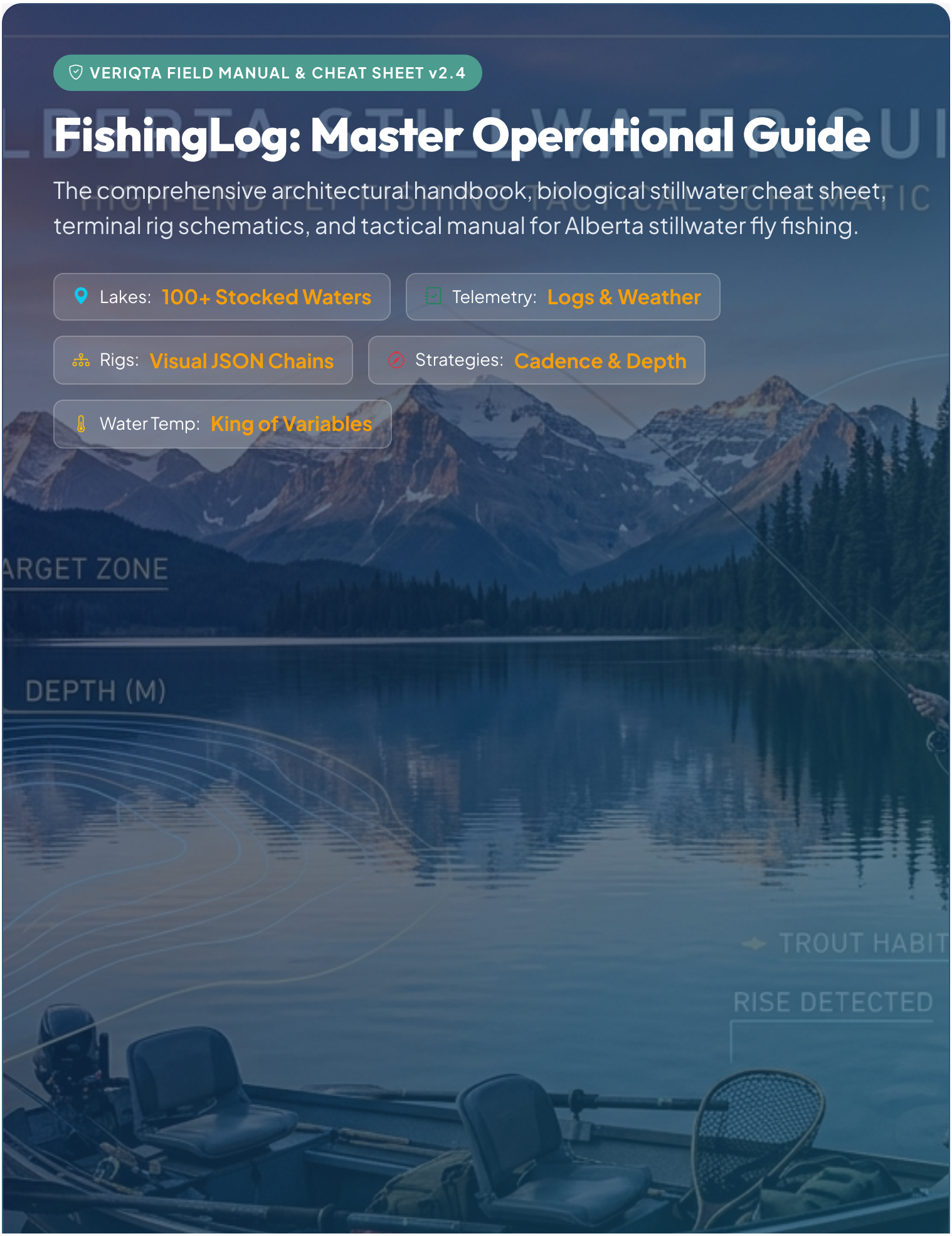
VERIQTA FIELD MANUAL & CHEAT SHEET v2.4

FishingLog: Master Operational Guide

The comprehensive architectural handbook, biological stillwater cheat sheet, terminal rig schematics, and tactical manual for Alberta stillwater fly fishing.

- Lakes: **100+ Stocked Waters**
- Telemetry: **Logs & Weather**
- Rigs: **Visual JSON Chains**
- Strategies: **Cadence & Depth**
- Water Temp: **King of Variables**



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Section 1: The 3x3 Core Architecture & Data Loop

SYSTEM TOPOLOGY

FishingLog is engineered around an integrated **3×3 Grid Ecosystem** where biological science, terminal tackle architecture, spatial lake telemetry, and angler catch logs form a self-reinforcing intelligence loop.

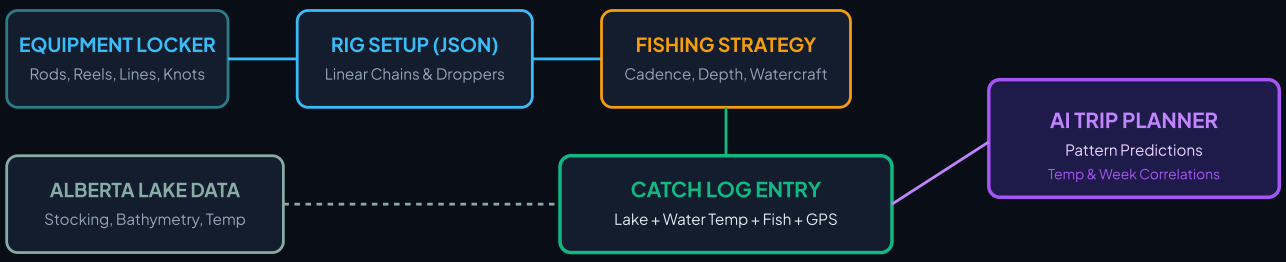


Figure 1.1: The Closed-Loop Telemetry Pipeline — How Rigging, Strategy, Lake Data, and Catches Feed the Predictive Engine.

1

Row 1: Biological & Spatial

Lakes: Alberta fisheries, bathymetric PDFs, regulations, GPS, automated distance calculations.
Bugs: Entomology, larval/pupal life cycles, and temperature triggers.
Flies: Pattern imitations, hook sizes, bead weight formulas, color schemes.

2

Row 2: Mechanical & Rigging

Equipment: Rods, reels, fly line sink rates, fluorocarbon/nylon tippets, terminal hardware.
Knots: Knot-tying strength ratings, loop mechanics, step-by-step knot cards.
Setups: Interactive visual JSON rig sequence builder with multi-dropper branches.

3

Row 3: Strategy & Community

Fish: Alberta species profiles, strain genetics, ATS triploid stocking numbers.
Strategies: Linking a physical rig to retrieval cadence, depth zone, and drift tactics.
Blog: Detailed field reports, technique breakdowns, and community updates.

> 💡 **Core System Rule:** Always build your **Locker Equipment** and **Knots** first. These components assemble into your **Setups (Rigs)**. Your Setups attach to **Strategies**. When you log a catch on desktop or mobile, picking a Strategy auto-loads its rig, logging a complete snapshot of how that fish was fooled!

Section 2: "Where Water Temp is King" — Stillwater Limnology

BIOLOGICAL SCIENCE

In Alberta's pothole and prairie trout lakes, water temperature is the master variable. It dictates fish metabolism, dissolved oxygen concentrations, aquatic insect metamorphosis, and where trout feed throughout the day.

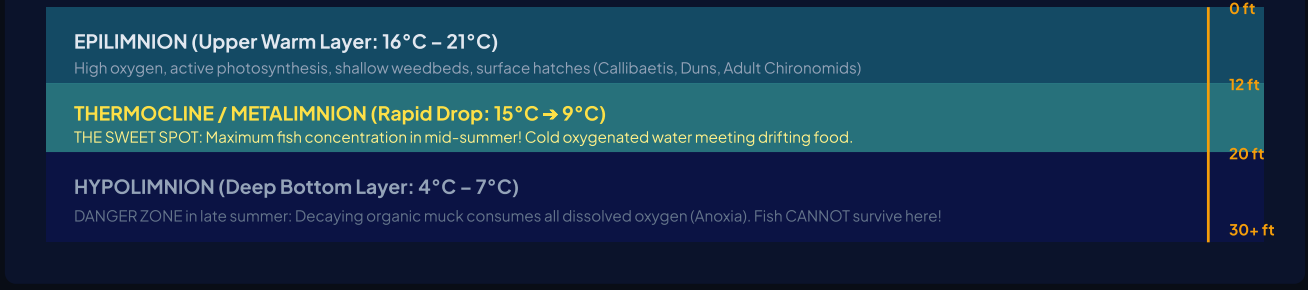


Figure 2.1: Summer Lake Thermal Stratification Profile — Target the Thermocline Sweet Spot.

Alberta Water Temperature Tactical Reference

TEMP RANGE	SEASONAL PHASE	TROUT BEHAVIOR & METABOLISM	DOMINANT FOOD TRIGGERS	OPTIMAL TACTIC
0°C – 4°C	**Ice-Out Turnover**	Sluggish metabolism. Fish hug shallow, dark mud bays that warm 1–2 degrees in midday sun.	Chironomid bloodworms (larvae), micro leeches, dormant scuds.	Slow bottom crawl, unweighted leech or bloodworm on long fluorocarbon leader.
4°C – 10°C	**Early Spring Feed**	Rapidly rising metabolic rate. Trout aggressively patrol warming shoals and drop-off shelves (6–12 ft).	First chironomid pupae ascents, water boatman migration, active scuds.	Suspended chironomid pupa under breakaway indicator 12 inches off bottom.

TEMP RANGE	SEASONAL PHASE	TROUT BEHAVIOR & METABOLISM	DOMINANT FOOD TRIGGERS	OPTIMAL TACTIC
10°C – 16°C	**PRIME WINDOW (Golden Era)**	Optimal metabolic efficiency. Heavy multi-species hatches; intense feeding across all depths.	Massive chironomid bombers (#10–#12), Callibaetis mayflies, damselfly crawls.	Dual dropper indicator rigs or Type 5 sinking line stripped along weedlines.
16°C – 19°C	**Summer Stratification**	Surface waters warm up. Trout retreat to the **Thermocline** (12–22 ft) or weedbed shade during midday.	Deep chironomids, dragonfly nymphs, evening travelling sedges (caddis).	Deep hanging indicator rigs (18–24 ft) or sinking line hanging techniques.
20°C+	**CRITICAL STRESS**	Lethal oxygen depletion. Fish cannot metabolize lactic acid. Prolonged fights cause high mortality.	Minimal daytime feeding; brief surface activity at dawn/dusk.	**SUSPEND FISHING.** Wait for cooler weather or fish early mornings (6:00 AM - 9:00 AM) only.

⚠️ **Field Gotcha:** Never fish directly on the bottom in late July or August in deeper lakes! Deep hypolimnion water can have zero dissolved oxygen. Your indicator rig must suspend your flies above the anoxic zone right at the edge of the thermocline.

Section 3: Alberta Lakes, Stocking Intelligence & Spatial GIS

FISHERIES GIS

The FishingLog lake registry synchronizes Alberta Environment & Protected Areas (AEP) stocking data, regional boundaries, GIS coordinates, bathymetric contours, and driving distances from your home profile.



Mitchel Lake
Central Parkland



Cardiff Pond
Edmonton Metro



Rainbow Trout
Fraser / Beaty / ATS



Tiger Trout
Piscivorous Hybrid



Brown Trout
Weed & Night Feeder



Brook Trout
Cold Water Char

Deciphering Alberta Stocking Records & Strain Codes

STRAIN CODE	BIOLOGICAL CLASSIFICATION	TROPHY GROWTH & FEEDING CHARACTERISTICS	ANGLING IMPLICATIONS
RNTR (ATS)	**All-Female Triploid Rainbow Trout**	Genetically modified with 3 sets of chromosomes (3N). 100% sterile; never expend energy developing eggs or spawning. Continues aggressive feeding straight through spring.	Grows to massive trophy sizes (5–10+ lbs) in rich pothole lakes. Hits flies hard even when standard fish are spawning.

STRAIN CODE	BIOLOGICAL CLASSIFICATION	TROPHY GROWTH & FEEDING CHARACTERISTICS	ANGLING IMPLICATIONS
TGTR	**Tiger Trout (Hybrid)**	Sterile intergeneric cross between Female Brown Trout (*Salmo trutta*) and Male Brook Trout (*Salvelinus fontinalis*).	Fiercely predatory. Feeds primarily on forage fish, minnows, leeches, and large dragonfly nymphs. Best targeted with large streamers and strip-pause cadences.
BRNT	**Brown Trout**	Wild and stocked strains. Highly wary, structure-oriented, nocturnal feeding tendencies.	Target low-light dawn/dusk hours and deep weed edges. Exceptional predatory strikes on balanced leeches and floating mouse patterns.
BKTR	**Brook Trout**	Char species native to eastern North America. Thrives in cold, spring-fed Alberta waters.	Readily takes attractor flies, scuds, and micro nymphs. Highly vulnerable to bright beadhead flies in early spring and late autumn.

Section 4: Aquatic Entomology & Fly Matching Matrix

FOOD & FLIES

Over 90% of an Alberta stillwater trout's diet consists of underwater aquatic insects. Matching the life stage and water column position is 10× more important than matching hook size.



Dragonfly Nymph
Bottom Prowler



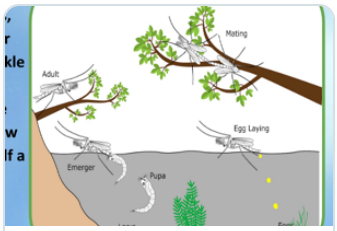
Callibaetis Nymph
Weedbed Swimmer



Damselfly Nymph
June Weedbed Crawl



Parachute Adams
Surface Dun / Adult



Chironomid Pupa
Primary Stillwater Prey



Bloodworm (Larva)
Benthic Stage (Hemoglobin)

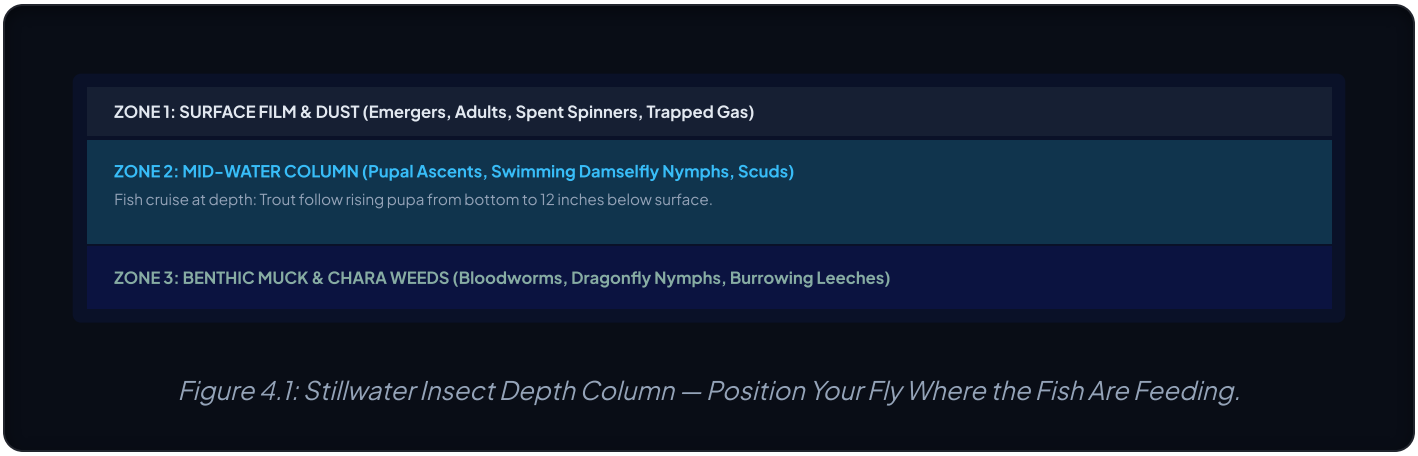


Figure 4.1: Stillwater Insect Depth Column — Position Your Fly Where the Fish Are Feeding.

The Big 6 Stillwater Food Organisms & Fly Recipe Cheatsheet

INSECT ORDER	PEAK SEASON	DISTINGUISHING HABITS	TOP FLY PATTERNS	SIZE & BEAD SPECS
Chironomid (Midge)	May 1 – July 15 & Sept	Pupae trap nitrogen gas, shining silver/chrome as they slowly swim vertically to surface.	Chromie, Ice Cream Cone, Anti-Static Buzzer, Zebra Midge, Red Bloodworm.	#10 - #18, White/Silver Tungsten or Glass Bead.
Callibaetis (Mayfly)	June – August	Speckled duns; nymphs swim with rapid abdominal flicks from weedbeds to surface.	Callibaetis Cripple, Hare's Ear, Parachute Adams, Pheasant Tail.	#12 - #16, 2X Fine wire, slender profile.
Damselfly	June 15 – July 20	Bright olive nymphs crawl from mud out to bulrushes to molt. S-swimming motion.	Livingston Damsel, Marabou Damsel, Red-Eyed Damsel, Floating Dragon.	#10 - #14 3X long hook, beadhead or monofilament eyes.
Dragonfly	All Season (Year-round)	Claw-legged bulky predators. Crawl slowly across bottom mud. Vulnerable when migrating.	Carey Special, Gomphus Nymph, Kaufmann's Dragon, Halfback.	#6 - #10 3X-4X long hook, heavy weighted body.
Leech	Ice-out to Freeze-up	Undulating horizontal ribbons. High-protein meal targeted at all depths.	Balanced Leech, Mohair Leech, Woolly Bugger, Simi Seal Leech.	#8 - #12, 60–degree jig hook with tungsten pin head.
Water Boatman & Backswimmer	Late August – October	Fall mating flight. Land on water with a distinct "splat", diving down with air bubbles.	Boatman Beetle, Prince Nymph, Foam Backswimmer, Silver-ribbed Hare's Ear.	#12 - #16, Pearlescent or silver tinsel air bubble belly.

Section 5: Equipment Locker & The Terminal Tackle Rig Builder

RIG ARCHITECTURE

The Rig Builder allows anglers to visually configure every element from fly rod to hook point. Setups are stored in structured JSON and resolve dynamic node cards and illustrated rig diagrams.

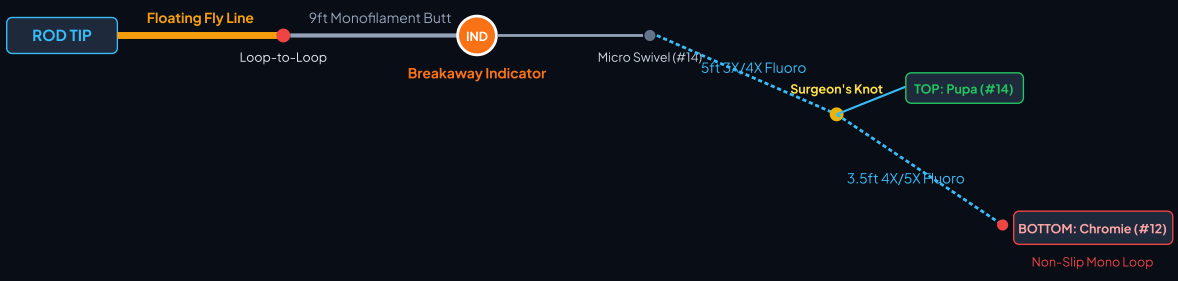


Figure 5.1: The Standard Dual-Fly Deep Stillwater Indicator Rig Architecture.

Equipment Locker Organization & Sort Orders

Order 1

Fly Rods

9'-10' 5wt-7wt

Order 2

Reels

Large Arbor Disc

Order 3

Fly Lines

Float / Sink Rates

Order 4

Leaders

9'-15' Monofilament

Order 5

Tippets

2X-6X Fluorocarbon

Order 6

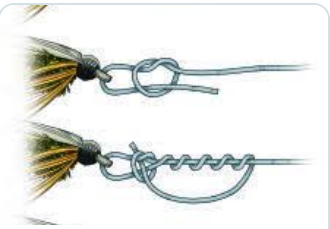
Hardware

Swivels, Rings, Snaps

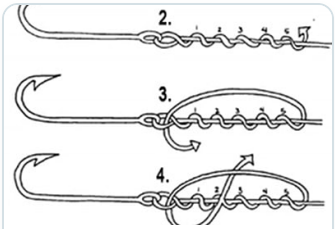
Section 6: Essential Fishing Knots & Strength Matrix

TERMINAL CONNECTIONS

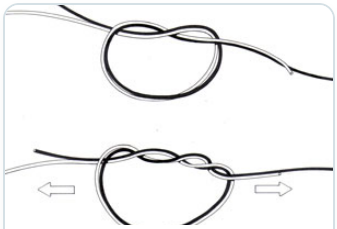
A trophy trout lost is almost always a knot failure. The Knot directory provides illustrated tying instructions, optimal line types, and relative breaking strength percentages.



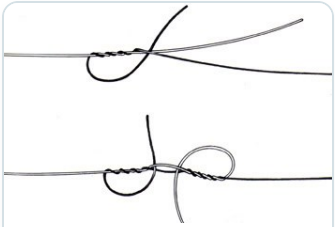
Non-Slip Mono Loop
Hook Freedom (95%)



Improved Clinch
Standard Eye (85%)



Double Surgeon's
Leader-to-Tippet (90%)



Blood Knot
Smooth Cast Link (88%)

KNOT NAME	STRENGTH %	PRIMARY APPLICATION	TYING MECHANICS & KEY GOTCHA	BEST LINE MATERIAL
Non-Slip Mono Loop	95%	Attaching nymphs, balanced leeches, and streamers to allow horizontal swimming action.	Make overhand knot in line, pass through eye, pass through overhand loop, wrap 4–5 times around standing line, pass back through loop. **Moisten with saliva before cinching!**	Fluorocarbon or Monofilament
Double Surgeon's Knot	92%	Joining leader to tippet; building dropper tags for 2-fly buzzer rigs.	Overlap leader and tippet by 6 inches. Form a loop and pass both strands through twice. Cut the upward tag for a clean, stiff dropper arm.	Equal or unequal diameter lines
Improved Clinch Knot	85%	Standard direct hook connection for dry flies, swivels, and weighted eyes.	5–7 turns around standing line. Thread tag through loop next to eye, then through big secondary loop. Pull slow and tight.	Monofilament nylon (Avoid with heavy fluoro)

KNOT NAME	STRENGTH %	PRIMARY APPLICATION	TYING MECHANICS & KEY GOTCHA	BEST LINE MATERIAL
Perfection Loop	90%	Butt section loops to connect leader to fly line welded loop.	Two counter-rotational loops, slip working line between loops, pull loop 2 through loop 1. Straight in-line pull.	Heavy butt monofilament (20lb-30lb)
Blood Knot	88%	Tying tapered leader sections together with zero bulk to glide through guides.	Wrap tag 1 around line 2 five times; wrap tag 2 opposite way five times; thread tags through center in opposing directions.	Strictly similar diameter lines (± 0.002")

Section 7: Fishing Strategies & Retrieval Cadence

TACTICAL EXECUTION

In FishingLog, a **Strategy** represents the tactical discipline applied on the water. It pairs an exact physical Rig Setup with specific retrieval cadence, boat drift speed, depth targeting, and weather compensation.

A

The Chironomid Dead-Drift

Cadence: 0 strips. Wind chop imparts subtle vertical bobbing through the indicator.
Adjustment: 1 slow hand-twist every 45–60 seconds to ascend the pupa 2 inches.
Depth: Suspended exactly 12 inches to 18 inches above the bottom weeds.

B

The Damselfly Pulse

Cadence: 3–4 inch smooth pulls with distinct 2–second pauses.
Adjustment: During June migrations, point rod tip toward the weedbed and let the fly undulate horizontally.
Line: Intermediate or Floating line with 12ft fluorocarbon leader.

C

The Sinking Line Deep Hang

Cadence: Fast 12–inch erratic strips along the drop-off, then a 10–second complete dead pause at the boat.
Trigger: 50% of aggressive strikes occur on the "hang" right as the line approaches vertical!

D

The Fall Boatman Twitch

Cadence: Sharp, erratic 1–inch twitches with rapid 3–inch dives.
Context: On bright September afternoons when boatmen dive from surface to weeds with trapped silver air bubbles.

Section 8: Catch Logging — Full Desktop vs. Mobile Quick Entry

TELEMETRY CAPTURE

Logging catches is the heart of the engine. FishingLog offers two synchronized interfaces: a comprehensive desktop journal for post-trip debriefs and an ultra-fast mobile interface designed for wet hands in a rocking boat.

Desktop Deep Log

- **Complete Measurement Telemetry:** Enter length in cm (auto-converts to inches) and weight in kg (auto-converts to lbs).
- **Depth Telemetry:** Record both *Total Lake Depth* (e.g. 18 ft) and exact *Catch Depth* (e.g. 16.5 ft) to calculate feeding offset.
- **Rich CKEditor Notes:** Full formatting for detailed observations, water clarity notes, and companion angler reports.
- **Photo Uploads & Tagging:** Cross-references the catch to your profile, lake, and fly pattern.
- **Privacy Toggle:** Mark logs private if you want to keep competition spots or confidential patterns secret.

Mobile Quick-Log

- **30-Second Touch Workflow:** Massive touch targets designed for cold, wet fingers on float tubes or boats.
- **Instant GPS Coordinates:** 1-tap geolocation button fetches your exact boat location instantly.
- **Live Weather Auto-Fill:** Automatically queries weather station for air temp, wind speed/direction, and barometric pressure.
- **Stocked Species Quick-Picker:** Selecting your lake automatically displays clickable chips for only the fish stocked in that waterbody.
- **Instant Strategy & Rig Linking:** Pick your active strategy in 1 tap; rig and fly defaults are applied instantly.

Section 9: Trip Planning & AI Pattern Engine

PREDICTIVE ANALYTICS

The FishingLog planning engine answers the ultimate question: *"Given current lake conditions and the date, where should I fish, at what depth, and with what fly?"*

1

Lake & Week Planning

The system pulls historical average water temperatures for that week, cross-references active hatches recorded across Alberta for that temperature, and shows what flies and depths yielded the highest historical catch rates!

2

AI Temperature & Date Forecast Engine

Enter current surface water temperature and date. The AI forecast engine computes insect emergence probabilities, determines whether the lake is turning over, stratified, or in the golden window, and generates a structured tactical flight plan for your day on the water.

Section 10: Ultimate Stillwater Quick-Reference Cheat Matrix

QUICK REFERENCE

Fly Line Sink Rate Classifications

LINE TYPE	SINK RATE	OPTIMAL TARGET DEPTH
Floating Line	0 ips (Floats)	0 – 20 ft (Under Indicator)
Hover / Clear Camo	1.0 – 1.25 ips	2 – 6 ft (Shallow Weedbeds)
Intermediate	1.5 – 2.0 ips	4 – 10 ft (Damsel & Mayfly)
Type 3 Full Sink	3.0 – 3.5 ips	8 – 15 ft (Mid-depth Drop-offs)
Type 5 Full Sink	5.0 – 5.5 ips	14 – 22 ft (Thermocline Stripping)
Type 7 Full Sink	7.0 – 8.0 ips	20 – 35+ ft (Deep Trench & Dragon)

The Tippet "Rule of 3" & Breaking Strengths

HOOK SIZE	FORMULA (÷ 3)	TIPPET (X)	AVG STRENGTH
#6 – #8 (Large Leech/Dragon)	8 ÷ 3 ≈ 2.6	**2X – 3X**	10.0 – 12.0 lbs
#10 – #12 (Chironomid Bomber)	12 ÷ 3 = 4.0	**4X**	7.5 – 8.5 lbs
#14 – #16 (Mayfly / Damsel)	15 ÷ 3 = 5.0	**4X – 5X**	5.5 – 6.5 lbs
#18 – #20 (Micro Midge)	18 ÷ 3 = 6.0	**6X**	3.5 – 4.5 lbs

Rule of 3 Formula: Divide fly hook size by 3 to find your matching tippet X-rating. (e.g. Size 12 fly ÷ 3 = 4X tippet)!

Alberta Stillwater Fisheries Species Code Index

RNTR = Rainbow Trout | TGTR = Tiger Trout | BRNT = Brown Trout | BKTR = Brook Trout | CTTR = Cutthroat Trout

LKTR = Lake Trout | ATS = All-Female Triploid | 2N / 3N = Diploid / Triploid